

Bi-Bridge: Bidirectional Diffusion Bridges for Low-Light Image Enhancement

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Abstract

*Low-Light Image Enhancement (LLIE) is a challenging task, as severe information loss means a single input can correspond to multiple plausible restorations. This inherent ambiguity causes conventional regression-based models to produce overly-smooth results that lack detail. While recent generative models can create richer details, their common unidirectional design often compromises content fidelity by distorting original structures. We introduce **Bi-Bridge**, a unified framework that models both enhancement and its inverse degradation within a single symmetric diffusion bridge. By compelling the network to preserve essential content structures across both transformations, this bidirectional learning acts as a powerful constraint, leading to significantly more faithful and realistic restorations. Extensive experiments show that Bi-Bridge outperforms state-of-the-art (SOTA) methods across multiple benchmarks, establishing a new standard for fidelity and perceptual quality.*

1. Introduction

The fundamental challenge in Low-Light Image Enhancement (LLIE) stems from the significant information loss inherent to low-light imaging. In dark or noisy regions, fine-grained textural and color information is often compressed or completely lost. This information loss gives rise to an inherent ambiguity: a single low-light image can correspond to multiple plausible normal-light counterparts, each with different yet valid details. This one-to-many relationship defines LLIE as a classic ill-posed inverse problem [22, 29] and poses a significant challenge for conventional models. As illustrated in Figure 1(a), end-to-end regression-based models like CNNs or Transformers [2, 34, 40], which are designed for one-to-one mappings, struggle with this ambiguity. Lacking an intrinsic mechanism to model the space of

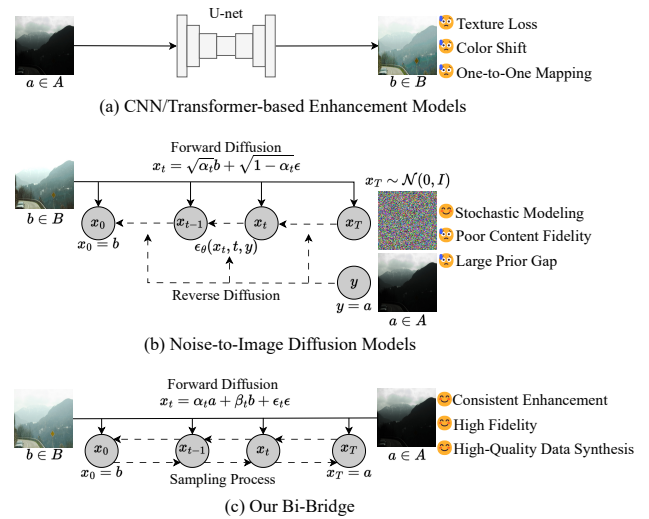


Figure 1. Conceptual comparison of LLIE paradigms and their fidelity challenges. (a) Regression-based models produce “averaged” results, leading to a loss of textural fidelity. (b) “Noise-to-Image” diffusion models start from a random prior, challenging content fidelity. (c) Our **Bi-Bridge** enforces bidirectional consistency, fundamentally enhancing restoration fidelity.

possible outputs, they often learn to produce an “averaged” prediction of all plausible solutions [4, 32], which tends to yield overly smooth textures and a loss of high-fidelity details.

Generative models, particularly diffusion models [8, 23], offer a powerful paradigm by modeling the distribution of possible restorations. However, the dominant “Noise-to-Image” framework (Figure 1(b)), used in works like [9, 10, 15], faces its own challenges. Starting from a random prior requires bridging a large domain gap, making it difficult to preserve the structural integrity of the source image. To mitigate this, recent works have introduced “Image-to-Image” diffusion bridge models like DDBM [46], which learn a direct mapping between data domains. While this reduces the domain gap, these bridge models have so far been predominantly explored in an asymmetric, unidirectional manner.

More profoundly, we observe that deep learning paradigms for LLIE predominantly adopt an asymmetric,

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unidirectional formulation, focusing solely on the restoration process. This observation prompts a central question: *Could jointly learning the degradation process with enhancement provide a powerful constraint for improving restoration fidelity?* We hypothesize that this symmetric training imposes a stronger constraint that mitigates the fidelity bottleneck seen in unidirectional approaches. To investigate this hypothesis, we introduce **Bi-Bridge**, a novel bidirectional diffusion bridge model designed explicitly to enforce this symmetric learning, as illustrated in Figure 1(c). The superior performance of our model, demonstrated in our experiments, provides strong evidence supporting this hypothesis.

Our approach materializes this hypothesis through a symmetric framework built upon DDBM. The technical core of our work lies in a simple yet powerful insight: the mean of DDBM’s tractable Gaussian bridge distribution is structurally symmetric with respect to its endpoints, $\mathbf{x}_0$ and $\mathbf{x}_T$. We exploit this inherent property to enforce bidirectional consistency with a single, unified U-Net predictor, D_θ . Specifically, for each training pair, we stochastically swap the roles of the startpoint $\mathbf{x}_0$ and endpoint $\mathbf{x}_T$. The network is then trained via a simple regression objective to predict the correct startpoint conditioned on the given endpoint. This design compels the network to learn a unified mapping that is invariant to the process direction, acting as a powerful implicit regularizer that significantly enhances structural fidelity and content preservation.

In summary, our main contributions are:

- We propose **Bi-Bridge**, a framework that introduces a bidirectional consistency constraint into the diffusion bridge paradigm to alleviate limitations of asymmetric LLIE models.
- We demonstrate that this symmetric training acts as a powerful implicit regularizer, significantly enhancing structural fidelity and content preservation.
- Our method achieves superior performance compared to state-of-the-art (SOTA) approaches across multiple datasets, setting a new benchmark for fidelity and perceptual quality in LLIE.

2. Related Work

2.1. Low-Light Image Enhancement Paradigms

Prevailing paradigms in LLIE offer different trade-offs in handling the task’s ill-posed nature. The first major approach consists of end-to-end regression-based models, including various CNN and Transformer architectures [2, 31, 33, 34, 39]. By learning a direct, one-to-one mapping, these methods often excel at preserving global structure. However, in facing the one-to-many nature of the problem, this approach can tend to produce “averaged” predictions, which may lead to overly-smooth textures and a compro-

mise in high-frequency detail fidelity [1].

A second paradigm of generative models, particularly diffusion models, offers an alternative by learning the distribution of possible restorations. The prevalent “Noise-to-Image” framework, adopted by methods such as [9, 10, 15, 38], can generate rich and diverse details. However, by starting from an uninformative random prior, these models face the challenge of preserving the precise content and structure of the source image across a large domain gap, which can sometimes affect content fidelity.

In contrast, our proposed **Bi-Bridge** is designed to mitigate challenges from both paradigms. As a generative model, it avoids the “averaged” results of regression methods. Simultaneously, by learning a direct bridge between the data domains, it mitigates the large domain gap issue of “Noise-to-Image” approaches, thereby better preserving content fidelity.

2.2. Bidirectionality in Generative Models

The aforementioned trade-offs have motivated exploring bidirectional formulations for a better balance between realism and fidelity. One such direction involves invertible models based on Normalizing Flows (NFs) [21], such as LLFlow [29]. While also addressing the ill-posed nature, invertibility in NF-based models is primarily a mathematical tool for exact likelihood training; the model is not explicitly trained to learn the physical degradation process. Concurrently, general-purpose bidirectional bridge frameworks like BDBM [14] have been proposed. Built upon the general mathematical theory of the Chapman-Kolmogorov Equation [13], BDBM offers a theoretically comprehensive framework for arbitrary data distributions. However, this very generality means it is not specifically optimized for vision tasks. Image restoration particularly benefits from strong structural priors to preserve high-frequency fidelity, a factor that a general-purpose design may not fully exploit.

In contrast, our work takes a more task-oriented approach. We introduce **Bi-Bridge** to improve upon the unidirectional DDBM [46] framework by enforcing a bidirectional consistency constraint. Exploiting the inherent structural symmetry of DDBM’s specific Gaussian bridge, our simpler design acts as a powerful regularizer, improving the balance between restoration fidelity and realistic detail generation.

3. Methodology

3.1. Preliminaries

Our work builds upon and extends the Denoising Diffusion Bridge Model (DDBM) [46], a powerful framework for paired image translation that improves upon standard diffusion models. Standard diffusion models [8] typically learn to reverse a process that corrupts data into pure Gaussian

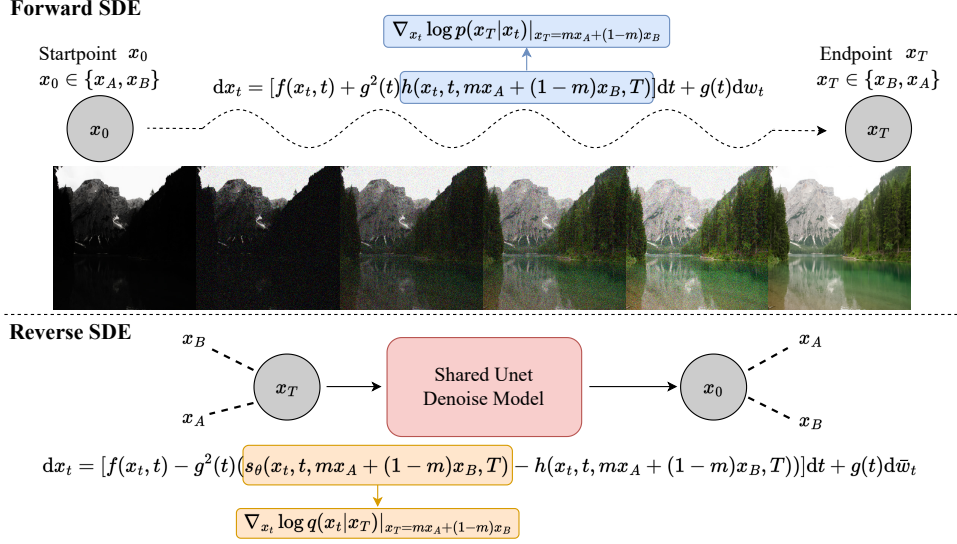


Figure 2. **Overview of the proposed Bi-Bridge architecture.** Our model utilizes a diffusion process guided by an analytical drift adjustment (in blue) towards a stochastically assigned endpoint $x_T \in \{x_A, x_B\}$. A shared U-Net learns to reverse this process by matching the denoising bridge score (in yellow). Conditioned on a direction indicator m , this allows a single reverse path to unify both image enhancement ($x_A \rightarrow x_B$) and degradation ($x_B \rightarrow x_A$).

noise. However, this “Noise-to-Image” paradigm is suboptimal for paired tasks like LLIE, which are better framed as a translation between two complex data distributions.

DDBM address this by learning the reverse of a conditioned stochastic process that forms a “bridge” directly between a source distribution (e.g., normal-light images $\mathbf{x}_0$) and a target distribution (e.g., low-light images $\mathbf{x}_T$).

Forward Bridge Process. The forward process in DDBM is a guided diffusion. It is defined by a Stochastic Differential Equation (SDE) that incorporates a Doob’s h-transform [5]. This transform introduces a guidance term $h(\cdot)$ that ensures the process starting from $\mathbf{x}_0$ precisely reaches the desired endpoint $\mathbf{x}_T$:

$$d\mathbf{x}_t = [f(\mathbf{x}_t, t) + g^2(t)h(\mathbf{x}_t, t, \mathbf{x}_T, T)] dt + g(t)d\mathbf{w}_t, \quad (1)$$

where $f(\cdot, \cdot)$ is the drift function, $g(\cdot)$ is the diffusion coefficient, and $\mathbf{w}_t$ is a standard Wiener process. The term $h(\cdot) = \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_T | \mathbf{x}_t)$ is the guidance term, representing the score of the transition probability to the endpoint.

Reverse Bridge Process and Learning. The key innovation of DDBM is to learn the corresponding time-reversal of this guided forward bridge. The reverse SDE is given by:

$$d\mathbf{x}_t = [f(\mathbf{x}_t, t) - g^2(t)(s(\mathbf{x}_t, \mathbf{x}_T, t) - h(\mathbf{x}_t, t, \mathbf{x}_T, T))] dt + g(t)d\bar{\mathbf{w}}_t. \quad (2)$$

The core learning task is to approximate the score function of the reverse process, $s(\mathbf{x}_t, \mathbf{x}_T, t) = \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_T)$. Since this score is intractable, DDBM

leverages the Denoising Bridge Score Matching principle [24]. It trains a neural network s_θ to match the score of a tractable distribution: the conditional bridge distribution $q(\mathbf{x}_t | \mathbf{x}_0, \mathbf{x}_T)$, which is known analytically. The objective function is:

$$\mathcal{L}_{\text{DDBM}}(\theta) = \mathbb{E}_{\mathbf{x}_t, \mathbf{x}_0, \mathbf{x}_T, t} \left[w(t) \left\| s_\theta(\mathbf{x}_t, \mathbf{x}_T, t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0, \mathbf{x}_T) \right\|^2 \right], \quad (3)$$

where $w(t)$ is a time-dependent loss weighting function.

While a powerful framework, the inherent unidirectionality (learning a mapping from a fixed source $\mathbf{x}_0$ to a target $\mathbf{x}_T$) limits its applicability for problems where a symmetric relationship is beneficial, and thus motivates our bidirectional extension.

3.2. Bi-Bridge: A Symmetric Diffusion Bridge Model

The inherent unidirectionality of the DDBM framework presents a fundamental limitation for LLIE, necessitating two separate models for enhancement and degradation. This approach doubles computational costs and, more critically, limits restoration fidelity by preventing the model from learning the intrinsic symmetric relationship of illumination changes. To address this, we propose **Bi-Bridge**, a unified framework where a bidirectional consistency constraint is enabled by exploiting the inherent structural symmetry of DDBM’s Gaussian bridge formulation.

3.2.1. Symmetric Bridge Formulation

The tractable Gaussian bridge distribution $q(\mathbf{x}_t | \mathbf{x}_0, \mathbf{x}_T)$ is defined as $\mathcal{N}(\mathbf{x}_t; \hat{\mu}_t, \hat{\sigma}_t^2 \mathbf{I})$, with the mean and variance

given by:

$$\hat{\mu}_t = \alpha_t \left(1 - \frac{\text{SNR}_T}{\text{SNR}_t}\right) \mathbf{x}_0 + \frac{\alpha_t}{\alpha_T} \frac{\text{SNR}_T}{\text{SNR}_t} \mathbf{x}_T, \quad (4)$$

$$\hat{\sigma}_t^2 = \sigma_t^2 \left(1 - \frac{\text{SNR}_T}{\text{SNR}_t}\right), \quad (5)$$

where α_t and σ_t are predefined signal and noise schedules, respectively, and $\text{SNR}_t = \alpha_t^2 / \sigma_t^2$ denotes the signal-to-noise ratio at time t .

Crucially, the mean $\hat{\mu}_t$ exhibits an explicit symmetric relationship with respect to the endpoints $(\mathbf{x}_0, \mathbf{x}_T)$. This inherent coupling allows us to model both the enhancement and degradation processes with a single, shared network. To implement this, for each training pair $(\mathbf{x}_A, \mathbf{x}_B)$, we stochastically determine the direction by sampling a binary variable m , which dynamically assigns the roles of the startpoint $\mathbf{x}_0$ and endpoint $\mathbf{x}_T$:

$$(\mathbf{x}_0, \mathbf{x}_T) = \begin{cases} (\mathbf{x}_A, \mathbf{x}_B), & \text{if } m = 0, \\ (\mathbf{x}_B, \mathbf{x}_A), & \text{if } m = 1. \end{cases} \quad (6)$$

3.2.2. Unified Training via Endpoint Prediction

While directly optimizing the score-matching objective is theoretically sound, prior work has shown that reparameterizing the objective to predict the startpoint $\mathbf{x}_0$ (the pred- x parameterization [12]) yields more stable training. We extend this parameterization to our bidirectional framework.

Instead of predicting the score, we train a single, unified network D_θ to predict the startpoint, $\hat{\mathbf{x}}_0 = D_\theta(\mathbf{x}_t, \mathbf{x}_T, t)$. The predicted startpoint $\hat{\mathbf{x}}_0$ can be used to reparameterize and approximate the score function. Following the formulation of DDBM, the intractable score $\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_T)$ is approximated by substituting the ground-truth startpoint $\mathbf{x}_0$ in the tractable score's formulation with our network's prediction $\hat{\mathbf{x}}_0$:

$$\begin{aligned} \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_T) &\approx s_\theta(\mathbf{x}_t, \mathbf{x}_T, t) = -\frac{\mathbf{x}_t - \hat{\mu}_t(\hat{\mathbf{x}}_0, \mathbf{x}_T)}{\hat{\sigma}_t^2} \\ &= -\frac{\mathbf{x}_t - \left(\alpha_t \left(1 - \frac{\text{SNR}_T}{\text{SNR}_t}\right) \hat{\mathbf{x}}_0 + \frac{\alpha_t}{\alpha_T} \frac{\text{SNR}_T}{\text{SNR}_t} \mathbf{x}_T\right)}{\sigma_t^2 \left(1 - \frac{\text{SNR}_T}{\text{SNR}_t}\right)}. \end{aligned} \quad (7)$$

This reparameterization demonstrates that training the network D_θ to predict $\mathbf{x}_0$ via a simple Mean Squared Error (MSE) loss is an effective and stable way to implicitly perform score matching. This leads to our final unified training objective:

$$\mathcal{L}_{Bi} = \mathbb{E}_{m, (\mathbf{x}_A, \mathbf{x}_B), t} [\|D_\theta(\mathbf{x}_t, \mathbf{x}_T, t) - \mathbf{x}_0\|^2], \quad (8)$$

where $\mathbf{x}_t$ is sampled from $q(\mathbf{x}_t | \mathbf{x}_0, \mathbf{x}_T)$. By minimizing this single objective, the shared predictor D_θ is forced to preserve invariant content while effectively modeling variable illumination, which is key to enhancing restoration fidelity.

3.3. Overall Training Objective

While the bridge loss $\mathcal{L}_{Bi}$ forms the core of our training, we incorporate an auxiliary perceptual loss [41], $\mathcal{L}_{per}$, to better align with human perceptual preferences. We employ this loss to enhance the learning of structure and texture details by penalizing the MSE between deep features of the prediction $\hat{\mathbf{x}}_0$ and ground truth $\mathbf{x}_0$:

$$\mathcal{L}_{per} = \|\phi(\hat{\mathbf{x}}_0) - \phi(\mathbf{x}_0)\|_2^2, \quad (9)$$

where ϕ denotes a feature extractor from a pre-trained network.

The final training objective is a weighted sum of these two components:

$$\mathcal{L} = \mathcal{L}_{Bi} + \lambda_p \mathcal{L}_{per}, \quad (10)$$

where λ_p is a hyperparameter that balances the two loss terms.

3.4. Bidirectional Inference

Once trained, the unified predictor D_θ can be seamlessly deployed for both enhancement and degradation. The inference process involves numerically solving the reverse SDE in Eq. (2) backward from $t = T$ to $t = 0$. The core of each step is approximating the SDE's drift term, which combines the analytical h-transform $h(\cdot)$ with the score function s_θ . This score function is dynamically derived at each step from our network's prediction, $\hat{\mathbf{x}}_0 = D_\theta(\mathbf{x}_t, \mathbf{x}_T, t)$, via the reparameterization in Eq. (7).

To ensure both high fidelity and efficient sampling, we adapt a higher-order hybrid sampler [12], as simple first-order solvers like Euler-Maruyama often require many steps. This predictor-corrector sampler first takes a stochastic step (e.g., Euler-Maruyama) to introduce beneficial noise for generating fine-grained textures, then performs a deterministic corrector step (e.g., Heun's method) for a more stable and accurate update. This combination effectively balances stochasticity with the stability of ODE-based solvers.

The specific application of this inference process is determined by the choice of the conditional endpoint $\mathbf{x}_T$. For enhancement $(\mathbf{x}_A \text{ to } \mathbf{x}_B)$, we set the endpoint $\mathbf{x}_T = \mathbf{x}_A$. The sampler is initialized at $t = T$ and proceeds backward to generate the normal-light image $\mathbf{x}_B$. Conversely, for degradation, we set $\mathbf{x}_T = \mathbf{x}_B$ to generate the low-light image $\mathbf{x}_A$. Detailed algorithms are provided in the supplementary material.

4. Experiments

4.1. Datasets

We evaluate our framework on a comprehensive suite of benchmark datasets. For experiments requiring paired

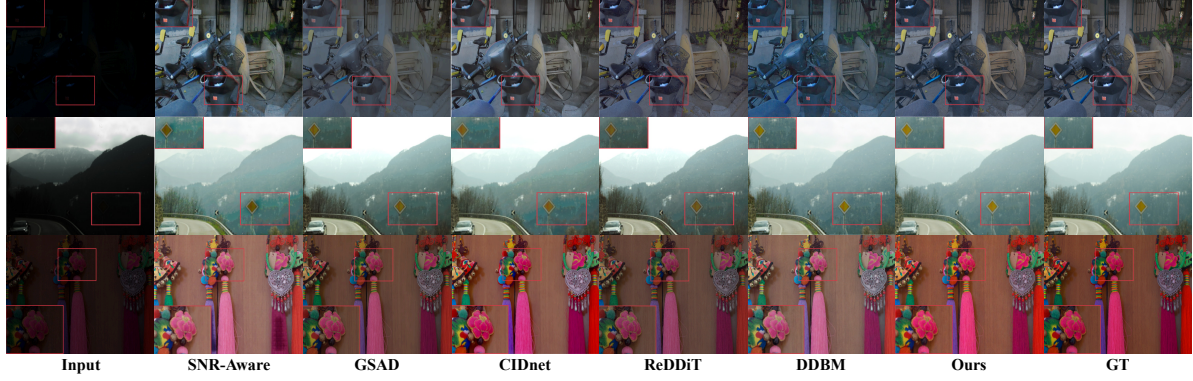


Figure 3. **Qualitative comparison with SOTA methods on paired datasets.** Results are shown for LOL-v2-real (top), LOL-v2-synthetic (middle), and LOL-v1 (bottom). Our model restores more natural illumination and finer textures, avoiding the artifacts and color shifts present in other methods, as highlighted in the red boxes. *Please zoom in for better visualization of details.*

Table 1. Evaluation results on LOL-v1 and LOL-v2(real and synthetic) datasets. The best performance is in **Red** color and the second best is in **Blue** color. Values in parentheses in the Ours row indicate the improvement over our unidirectional baseline, DDBM_(Baseline).

Methods	Source	LOL-v2-Synthetic			LOL-v2-Real			LOL-v1		
		PSNR (↑)	SSIM (↑)	LPIPS (↓)	PSNR (↑)	SSIM (↑)	LPIPS (↓)	PSNR (↑)	SSIM (↑)	LPIPS (↓)
Regression-based Methods										
LIME [7]	TIP'16	16.880	0.776	0.675	15.240	0.470	0.415	16.760	0.560	0.350
ZeroDCE [6]	CVPR'20	17.712	0.815	0.169	18.059	0.580	0.313	14.861	0.562	0.335
RetinexNet [31]	BMVC'18	17.130	0.754	0.754	18.371	0.723	0.365	16.770	0.462	0.474
KinD [42]	MM'19	16.259	0.591	0.435	17.544	0.669	0.375	20.870	0.799	0.207
KinD++ [43]	IJCV'20	-	-	-	19.087	0.817	0.180	21.300	0.823	0.175
DRBN [36]	CVPR'20	23.220	0.927	-	20.130	0.830	0.147	19.860	0.834	0.155
MIRNet [40]	TPAMI'22	21.940	0.846	-	20.357	0.782	0.317	24.140	0.842	0.131
SNR-Aware [34]	CVPR'22	27.787	0.941	0.054	27.209	0.871	0.157	26.716	0.851	0.152
LLFormer [27]	AAAI'23	24.807	0.919	0.067	26.200	0.888	0.137	25.132	0.872	0.117
Retinexformer [2]	ICCV'23	29.040	0.939	0.039	27.710	0.856	0.109	27.180	0.850	0.129
CIDNet [35]	CVPR'25	29.566	0.950	0.044	28.393	0.887	0.114	28.141	0.889	0.099
Noise-to-Image Diffusion Methods										
QuadPrior [28]	CVPR'24	16.108	0.758	0.114	20.592	0.811	0.202	22.849	0.800	0.201
PyDiff [45]	IJCAI'23	20.659	0.879	0.275	29.629	0.876	0.194	27.090	0.930	0.100
WCDM [10]	ToG'23	29.852	0.908	0.077	30.461	0.879	0.070	26.316	0.844	0.219
GSAD [9]	NIPS'23	28.670	0.944	0.047	28.818	0.895	0.095	27.839	0.877	0.091
ReDDiT [15]	CVPR'25	30.166	0.943	0.028	30.919	0.895	0.040	28.090	0.861	0.052
Image-to-Image Bridge Methods										
LLFlow [29]	AAAI'22	25.807	0.919	0.067	26.200	0.888	0.137	25.132	0.860	0.117
DDB [3]	NIPS'23	23.340	0.857	0.131	21.891	0.773	0.186	23.186	0.802	0.198
I ² SB [18]	ICML'23	26.155	0.879	0.089	23.959	0.794	0.198	23.659	0.791	0.172
BDBM [14]	KDD'25	28.053	0.885	0.075	25.102	0.811	0.158	25.567	0.820	0.161
DDBM _(Baseline) [46]	ICLR'24	27.872	0.881	0.079	26.353	0.813	0.165	24.451	0.812	0.159
Ours	-	31.019 _(+3.15)	0.944 _(+0.06)	0.025 _(-0.05)	31.287 _(+4.93)	0.841 _(+0.03)	0.040 _(-0.12)	27.879 _(+3.43)	0.865 _(+0.05)	0.052 _(-0.11)

ground truth, we utilize the cornerstone LOL dataset, including both LOL-v1 [31] (485 training pairs) and the more extensive LOL-v2 [37] (real and synthetic subsets). To assess generalization to real-world scenarios, we conduct zero-shot evaluation on five widely-used unpaired datasets: DICM [16], LIME [7], MEF [17], NPE [26], and VV [25], which collectively represent a diverse range of low-light conditions. Further implementation details for all experiments are provided in the supplementary material.

4.2. Evaluation Metrics

Our method's performance is assessed using widely-accepted full-reference and no-reference metrics. On paired datasets (LOL-v1/v2), we report Peak Signal-to-Noise Ra-

tio (PSNR) and Structural Similarity Index (SSIM) [30] to measure fidelity. To better align with human perception, we also report the Learned Perceptual Image Patch Similarity (LPIPS) [41]. For unpaired datasets, we assess perceptual quality using the Naturalness Image Quality Evaluator (NIQE) [20]. Higher PSNR/SSIM values are better, while lower LPIPS/NIQE values are better.

4.3. Comparison with State-of-the-Art Methods

As presented in Table 1, our proposed **Bi-Bridge** achieves SOTA performance, demonstrating a significant advantage over methods across all major LLIE paradigms, particularly in terms of PSNR and LPIPS. When compared to *Regression-based Methods*, **Bi-Bridge** surpasses top-

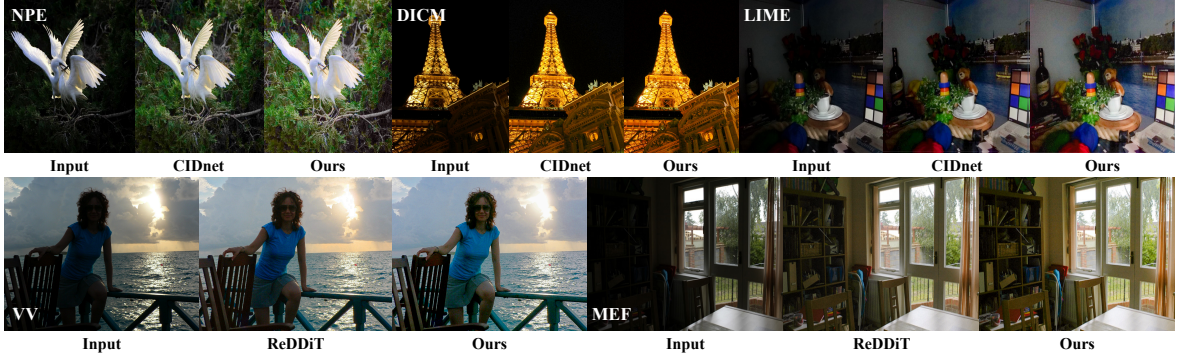


Figure 4. **Qualitative zero-shot generalization results on unpaired datasets.** Our method produces more natural illumination and realistic colors compared to prior SOTA approaches.

Table 2. Zero-shot generalization performance on unpaired datasets. We report NIQE scores ($\downarrow$). Best results are in **bold**.

Methods	DICM	LIME	MEF	NPE	VV
LIME [7]	5.42	3.84	3.73	4.37	3.63
Zero-DCE [6]	3.77	3.61	4.12	6.19	4.81
EnGAN [11]	3.98	4.49	4.95	3.42	3.93
KinD++ [43]	3.89	4.90	4.55	3.91	3.82
SCI [19]	3.73	3.52	4.07	6.44	4.05
DCC-Net [44]	3.84	6.27	6.04	7.97	4.66
Retinexformer [2]	4.47	3.96	4.12	4.28	4.53
WCDM [10]	6.38	4.62	4.86	4.65	5.39
GSAD [9]	4.35	3.97	4.08	5.35	3.18
CIDNet [35]	3.79	4.13	3.56	3.74	3.21
ReDDiT [15]	3.62	3.45	3.93	3.24	3.00
Ours	3.35	3.14	3.11	3.11	3.19

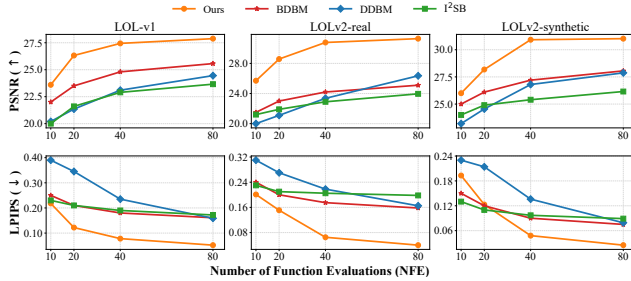


Figure 5. **Performance-efficiency comparison with other Image-to-Image Diffusion Bridge Models.** PSNR ($\uparrow$) and LPIPS ($\downarrow$) are plotted against NFE.

performing models like CIDNet by a significant margin (e.g., **+2.894** dB in PSNR on LOL-v2-real), showcasing the inherent advantage of our generative formulation. It also demonstrates a clear and consistent advantage over leading *Noise-to-Image Diffusion Methods* like ReDDiT, particularly in perceptual metrics.

Within the *Image-to-Image Bridge Methods* category, **Bi-Bridge** again shows superior performance. It significantly outperforms other bridge models, including the general-purpose bidirectional framework BDBM [14], which validates our design choices tailored specifically

for the LLIE task. Most critically, **Bi-Bridge** achieves a massive performance leap over its direct unidirectional baseline, $\text{DDBM}_{(\text{Baseline})}$, exemplified by a remarkable **+4.934** dB PSNR improvement on the challenging LOL-v2-real dataset. This dramatic gain, not attributable to any architectural changes, empirically validates our central claim: enforcing a symmetric consistency constraint acts as a powerful regularizer that fundamentally enhances restoration fidelity. A more detailed analysis of this comparison is provided in the Ablation Study (Section 4.6). The qualitative results in Figure 3 further corroborate these quantitative findings, where our model consistently restores more faithful colors and sharper details, avoiding the artifacts common in other SOTA methods.

4.4. Generalization on Unpaired Datasets

To validate the model’s generalization to real-world, unseen degradations, we conduct a zero-shot evaluation on five unpaired datasets using a model trained on LOL-v2-syn. As shown in Table 2, **Bi-Bridge** demonstrates remarkable robustness, achieving SOTA NIQE scores on four out of the five datasets. The qualitative results in Figure 4 further corroborate this strong performance. Our method consistently produces images with balanced contrast and natural colors, avoiding the under-enhancement or unnatural color casts common in competing methods. This consistent, top-tier performance without any fine-tuning highlights the robustness endowed by our symmetric learning paradigm.

4.5. Comparison with Other Image-to-Image Diffusion Methods

To specifically isolate the advantages of our bidirectional paradigm, we conduct a detailed performance-versus-efficiency analysis against other prominent *Image-to-Image Diffusion Bridge Models*. As illustrated in Figure 5, we plot the PSNR and LPIPS scores against the number of function evaluations (NFE). The results provide strong evidence for the significant benefits of our symmetric training objective across all sampling budgets.

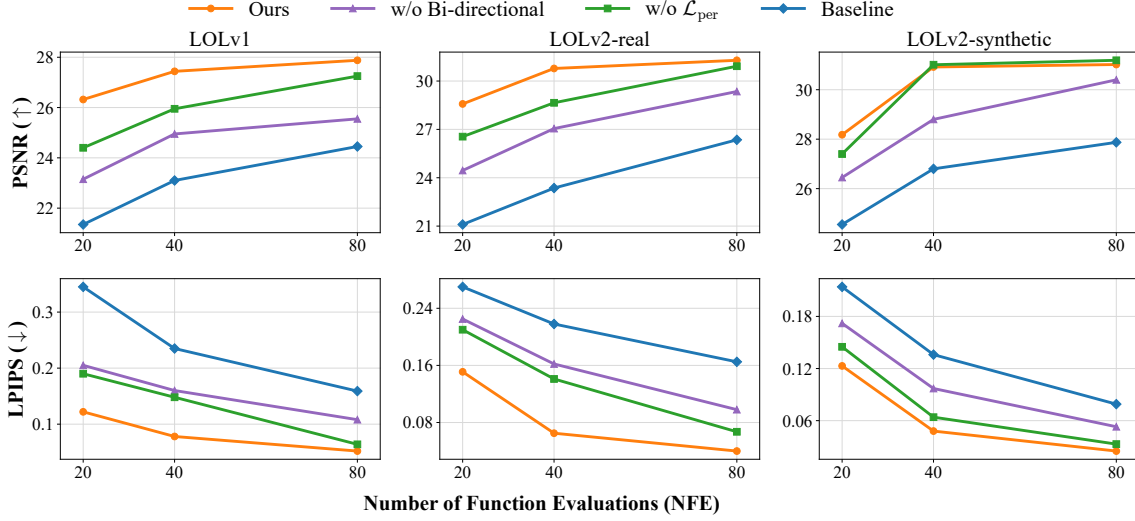


Figure 6. **Ablation study on our key components.** We compare our full model (*Ours*) against three ablated versions across varying NFEs.

Our **Bi-Bridge** (the orange curve) consistently establishes a new SOTA within this category, significantly outperforming all unidirectional counterparts, including its direct baseline DDBM, as well as other strong methods like I^2SB , BDBM. This highlights the fundamental advantage of enforcing bidirectional consistency.

More critically, our method exhibits significantly faster convergence and superior sampling efficiency. In terms of performance, **Bi-Bridge** with a minimal budget of just 10 NFEs consistently achieves results comparable to or even exceeding other bridge models operating at 80 NFEs. For instance, on the challenging LOL-v1 dataset, our 10-NFE model already surpasses the 80-NFE performance of both DDBM and I^2SB in PSNR. This superior efficiency, a direct consequence of the robust representation learned through bidirectional consistency, validates **Bi-Bridge** as not only a more accurate but also a significantly more practical solution for high-fidelity low-light enhancement.

4.6. Ablation Study

To validate the contribution of our key design components, we conduct a series of ablation studies, comparing our full model (*Ours*) against three ablated variants: 1) a *Baseline* unidirectional DDBM, 2) a model without our bidirectional training objective (*w/o Bi-directional*), and 3) a model without the auxiliary perceptual loss (*w/o $\mathcal{L}_{per}$*). Figure 6 reveals a consistent performance hierarchy, where our full model outperforms all ablated versions, demonstrating the complementary benefits of each component.

The most critical finding is the pivotal role of the bidirectional consistency constraint. The substantial margin between the *w/o Bi-directional* model and the *Baseline* already verifies the intrinsic advantage of the bridge-based formulation over standard unidirectional diffusion. More importantly, the subsequent sharp improvement achieved by our

full model empirically validates that enforcing symmetry is the primary driver of our success. This is attributable to its powerful regularization effect, as forcing the network to master both processes compels it to learn a more fundamental representation that distinguishes content from illumination. This prevents the overfitting to spurious correlations common in unidirectional training and leads to a model that is not only more accurate but also generalizes better.

Beyond superior accuracy, our symmetric design yields remarkable efficiency gains. As shown in Figure 6, our model converges faster. Notably, a 20-NFE **Bi-Bridge** already matches or surpasses the performance of the 80-NFE *Baseline* on LOLv2-real, reflecting an approximate $4\times$ improvement in sampling efficiency. This efficiency stems from the symmetric objective, which aligns both processes to learn a more direct enhancement path. The auxiliary perceptual loss further strengthens this alignment.

4.7. Data Synthesis Capability

An interesting emergent capability of our bidirectional framework is its function for data synthesis, offering a potential direction to mitigate the chronic data scarcity in LLIE. To evaluate this, we assess our model’s ability for high-fidelity degradation by conducting a challenging out-of-domain generalization test, applying our degradation model (trained on LOL-v2-syn) to novel high-resolution images. Lacking standard quantitative metrics, we rely on qualitative visualization.

A particularly noteworthy finding, visualized in Figure 7, is that our bidirectionally trained **Bi-Bridge** produces higher-fidelity degradations than a unidirectional DDBM baseline trained specifically for this task on the same data. This comparison reveals a surprising advantage of our symmetric training: **Bi-Bridge** excels even for the degradation task it was not exclusively optimized for.

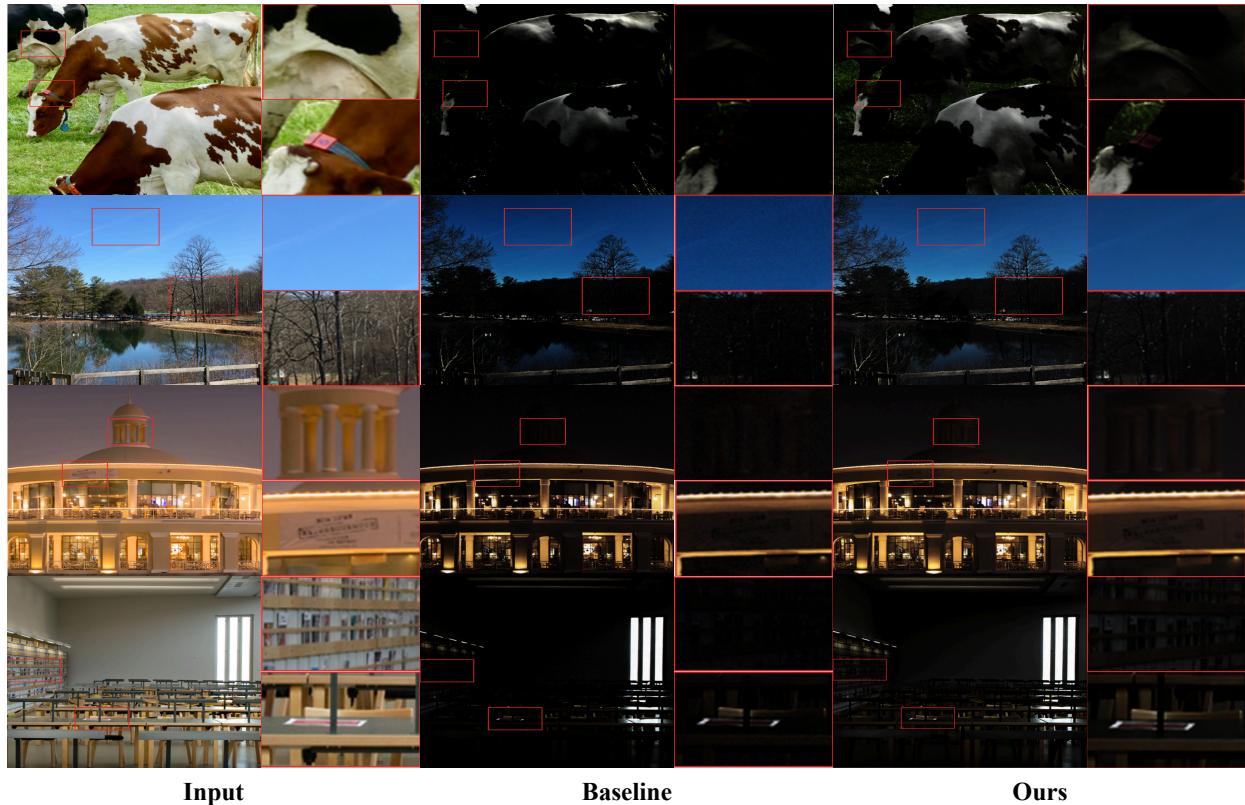


Figure 7. **Visualization of the data synthesis capability of Bi-Bridge.** From left to right: the original normal-light image (*Input*), the synthesized low-light counterpart from a unidirectional DDBM baseline trained only for degradation (*Baseline*), and the result from our bidirectionally trained **Bi-Bridge** (*Ours*). Please zoom in for a detailed view of the highlighted regions.

A detailed visual analysis in Figure 7 validates this superiority. In terms of detail preservation, our model consistently retains more subtle textures in extreme shadows. For example, in the cow scene (Row 1) and the library scene (Row 4), our method preserves fine details on the animal’s hide and the books on the shelves, whereas the baseline tends to crush these near-black regions into uniform dark patches. Furthermore, our model demonstrates more realistic color and texture rendition. In the lake landscape (Row 2), our approach renders a smooth, deep blue twilight sky with fine details in the treeline silhouette, while the baseline’s sky is noisy and its dark areas are devoid of detail. This fidelity extends to artificial lighting; in the rotunda scene (Row 3), our model correctly captures the warm color temperature and preserves the texture on the banner under the light, details that are either lost or rendered with less accuracy by the baseline.

This superior qualitative performance provides strong evidence that our symmetric training paradigm learns a more fundamental and transferable representation of illumination change. The fact that a bidirectionally trained model outperforms a specialist unidirectional model suggests that forcing the network to understand both forward and reverse

processes acts as a powerful regularizer, leading to more robust and physically-plausible generative capabilities. This highlights the immense potential of **Bi-Bridge** as a practical tool for high-fidelity data augmentation.

5. Conclusion

In this work, we revisited the asymmetric, unidirectional formulation of prevailing LLIE paradigms, a design choice that can compromise restoration fidelity. We introduced **Bi-Bridge**, a framework enforces bidirectional consistency within a symmetric diffusion bridge. This symmetric training acts as a powerful regularizer, leading to SOTA performance by compelling the network to learn a more robust representation of invariant content.

While our work sets a new benchmark for fidelity, we acknowledge that its iterative nature presents challenges for real-time applications, highlighting inference acceleration techniques as a key direction for future work. More profoundly, we believe **Bi-Bridge** marks a significant step towards a more holistic understanding of image formation by demonstrating the power of symmetric, bidirectional learning, paving the way for more robust and faithful image restoration models.

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